

Nitesh Upadhyay

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LinkedIn

GitHub

Portfolio

Summary

Generative AI Engineer with **2.6+ years** of experience building enterprise-grade AI applications using **Python, FastAPI, LangChain, LangGraph, MCP, RAG, and AWS Bedrock**. Skilled in developing **Agentic AI**, AI Agents, multimodal RAG, vector search, and scalable LLM applications with planning, tool calling, memory, and HITL. Experienced in AWS, Docker, REST APIs, observability, LLM evaluation, prompt engineering, and delivering secure, production-ready cloud-native AI solutions.

Technical Skills

Languages: Python, SQL

Backend: FastAPI, Django, Flask, REST APIs, JWT, RBAC

GenAI & Agentic AI: LangChain, LangGraph, MCP, CrewAI, AI Agents, Agentic AI, Multi-Agent Systems, A2A, RAG, Advanced RAG, Multimodal RAG, Prompt Engineering, Tool Calling, Function Calling, HITL, LLM Evaluation, Guardrails

LLMs: OpenAI, AWS Bedrock, Claude, Gemini, Llama, Hugging Face

Vector DB & Search: ChromaDB, FAISS, Pinecone, Semantic Search, Hybrid Search, Reranking

Cloud & DevOps: AWS (Bedrock, Lambda, S3, Textract, RDS, IAM, CloudWatch), Docker, Git, Azure DevOps, Linux

Databases: PostgreSQL, MySQL, MongoDB

Experience

Hexaware Technologies

Jan 2026 – Present

Generative AI Engineer

Noida, India

- **Project: Enterprise AI-Powered Bid Management System**
- Developed enterprise-grade **Agentic AI** solutions using **Python, FastAPI, LangChain, LangGraph, and AWS Bedrock**, implementing planning, reasoning, tool calling, memory, checkpointing, retries, state persistence, and Human-in-the-Loop (HITL).
- Built **MCP (Model Context Protocol)** servers and **Agent-to-Agent (A2A)** workflows. Developed advanced **RAG** systems using semantic search, hybrid retrieval, reranking, metadata filtering, and LLM evaluation.
- Designed multimodal document intelligence using **AWS Textract** and Bedrock for proposal generation, summarization, question answering, and workflow automation.
- Implemented Responsible AI, observability, monitoring, Docker, CI/CD, and cloud-native deployment using **AWS Bedrock, Lambda, S3, RDS, IAM, CloudWatch, and Azure DevOps**.

Tata Consultancy Services (TCS)

Jan 2024 – Dec 2025

Python Generative AI Developer

Nagpur, India

- **Project: Aptiv ADAS System Enhancement**
- Developed Generative AI solutions using the **RESIM** framework. Built **RAG** pipelines with LangChain, ChromaDB, and OpenAI, and implemented Agentic AI workflows using LangGraph and CrewAI.
- **Project: TCS iON Learning Platform**
- Integrated Generative AI using **Python, Flask, LangChain, and Google Gemini**, improving engagement by **90%**. Developed secure FastAPI/Flask APIs with JWT authentication and optimized backend performance, reducing development effort by **80%**.
- Built production-ready AI services using **Python, FastAPI, LangChain, OpenAI, Google Gemini, ChromaDB, Docker, Git, and AWS**, following Agile and enterprise backend best practices.

Projects

Enterprise Agentic AI Assistant | Python, LangGraph, LangChain, MCP, FastAPI, AWS Bedrock

- Developed an enterprise-grade **Agentic AI** assistant with autonomous planning, reasoning, memory, checkpointing, retries, and Human-in-the-Loop (HITL) approvals using LangGraph.
- Built **MCP (Model Context Protocol)** servers enabling AI agents to securely access enterprise APIs, databases, file systems, and third-party business applications.
- Implemented multi-agent collaboration with tool calling, structured outputs, state persistence, and Agent-to-Agent (A2A) communication for workflow automation.

- Deployed scalable FastAPI microservices on AWS using Bedrock, Lambda, S3, IAM, Docker, and CloudWatch with monitoring and observability.
- Multimodal RAG Knowledge Assistant | Python, LangChain, OpenAI, ChromaDB, FAISS, AWS Textract**
- Built an enterprise **Multimodal RAG** system supporting PDFs, Word documents, images, and scanned files using AWS Textract and vector databases.
 - Implemented semantic search, hybrid retrieval, metadata filtering, reranking, contextual compression, and citation generation for accurate knowledge retrieval.
 - Integrated LLM evaluation, hallucination detection, prompt engineering, and guardrails to improve groundedness and answer quality.
 - Developed secure FastAPI APIs with JWT authentication and Docker deployment for scalable document intelligence applications.
- AI Resume Screening & Recruitment Agent | Python, LangGraph, FastAPI, PostgreSQL, AWS Bedrock**
- Developed an AI recruitment agent capable of resume parsing, ATS scoring, semantic candidate matching, interview scheduling, and automated recruiter workflows.
 - Implemented Agentic AI workflows using LangGraph with planning, tool calling, memory, multi-step reasoning, and Human-in-the-Loop approvals.
 - Built RAG pipelines using vector embeddings for intelligent candidate-job matching and personalized interview question generation.
 - Integrated observability, prompt optimization, LLM evaluation, Docker, CI/CD, and AWS services for secure production deployment.

Education

Master of Computer Applications (CGPA: 8.25/10) Lakshmi Narain College of Technology (LNCT), Bhopal, MP	2021 – 2023
Bachelor of Science (PCM) – 68% Awadhesh Pratap Singh University (APSU), Rewa, MP	2018 – 2021
Higher Secondary (PCM) – 90% Madhya Pradesh Board of Secondary Education (MPBSE), Rewa, MP	2018

Achievements

- Won **1st Place** in the **TCS AI Fridays Hackathon** among 100+ teams for developing an innovative AI solution.
- Recognized as **Employee of the Month** for delivering high-impact Generative AI solutions and consistently exceeding project expectations.
- Received the **Most Innovative Idea Award** for designing an enterprise Agentic AI solution that improved workflow automation and productivity.
- Ranked in the **Top 1%** of the **M.P. Board (Science Stream)**, securing the 99th percentile statewide.
- Earned industry certifications including **AWS Certified Developer – Associate**, **AWS Certified Cloud Practitioner**, and **Microsoft Azure Data Fundamentals (DP-900)**.